MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guidelines for Treatment of Bacterial Meningitis in Pediatrics (Community- or Intrapartum-acquired)

<table>
<tr><th>Common Pathogens</th><th>Empiric Therapy</th><th>Duration of Therapy</th><th>Comments/Reference</th></tr>
<tr><td rowspan="4"><u><1 month*</u>

S. agalactiae,
E. coli,
L. monocytogenes,
Klebsiella spp</td><td><u>0-7 days (< 2000 g):</u>
Ampicillin 50 mg/kg/dose IV q12h
+ Cefotaxime 50 mg/kg/dose IV q12h</td><td rowspan="5"><u>If an organism is identified, the typical duration of therapy is:</u>

N. meningitides:
7 days

H. influenzae:
7 days

S. pneumoniae:
10-14 days

S. agalactiae:
14-21 days

Aerobic GNRs:
21 days

L. monocytogenes:
≥ 21 days</td><td rowspan="5">• CT prior to lumbar puncture if:
○ Immunocompromised
○ History of CNS disease (mass lesion, stroke)
○ New onset seizures
○ Papilledema
○ Abnormal level of consciousness
○ Focal neurologic deficit

• The use of dexamethasone has demonstrated a decrease in morbidity in children with meningitis due to H. influenzae type B and in adults with meningitis due to S. pneumoniae. Dexamethasone may be considered for pediatric patients with meningitis due to either H. influenzae or S. pneumoniae.

• Dexamethasone should be administered 10-20 min before antimicrobial therapy for maximal efficacy and continued for 2-4 days.

• If dexamethasone is utilized for meningitis, then rifampin (10-20 mg/kg/day, max 600 mg/day) should also be administered.

• If an organism is identified via CSF culture, then definitive therapy should be narrowed to a single agent with good CSF penetration and activity against the causative organism.

• If history or physical exam findings are concerning for HSV encephalitis, consider the use of acyclovir.</td></tr>
<tr><td><u>0-7 days (> 2000 g):</u>
Ampicillin 50 mg/kg/dose IV q8h
+ Cefotaxime 75 mg/kg/dose IV q12h</td></tr>
<tr><td><u>8 days-30 days (< 1200 g):</u>
Ampicillin 50 mg/kg/dose IV q12h
+ Cefotaxime 50 mg/kg/dose IV q12h</td></tr>
<tr><td><u>8 days-30 days (1200-2000 g):</u>
Ampicillin 50 mg/kg/day IV q8h
+ Cefotaxime 37.5 mg/kg/dose IV q6h or 50 mg/kg/dose IV q8h</td></tr>
<tr><td><u>8-30 days (> 2000 g):</u>
Ampicillin 50 mg/kg/dose IV q6h or 67 mg/kg/dose IV q8h
+ Cefotaxime 50 mg/kg/dose IV q6h or q8h</td></tr>
<tr><td><u>1 - 2 months</u>

S. pneumoniae,
N. meningitidis,
S. agalactiae,
H. influenzae</td><td>Cefotaxime 75 mg/kg/dose IV q6h or q8h
+ Vancomycin IV</td></tr>
<tr><td><u>2 months- 18 years</u>

N. meningitidis,
S. pneumoniae</td><td><u>Preferred:</u>
Ceftriaxone 50 mg/kg/dose IV q12h (max: 2 g/dose)
+ Vancomycin IV

<u>Non-life threatening penicillin or cephalosporin allergy:</u>
Meropenem 40 mg/kg/dose q8h (max: 2 g/dose)
+ Vancomycin IV

<u>Life threatening penicillin allergy:</u>
Aztreonam 40 mg/kg/dose q6h (max: 2 g/dose)
+ Vancomycin IV</td><td>If an organism is identified, the typical duration of therapy is:

N. meningitides:
7 days

S. pneumoniae:
10-14 days</td></tr>
</table>

Common Pathogens	Empiric Therapy	Duration of Therapy	Comments/Reference
Basilar skull fracture *S. pneumoniae*, *H. influenzae*, Group A strep	Preferred: **Ceftriaxone** 50 mg/kg/dose IV q12h (max: 2 g/dose) + **Vancomycin** IV Non-life threatening PCN/cephalosporin allergy: **Meropenem** 40 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Life threatening penicillin allergy: **Aztreonam** 40 mg/kg/dose IV q6h (max: 2 g/dose) + **Vancomycin** IV	7-21 days (plus)	• Patients with significant barrier disruption are at increased risk of resistant-gram negative organisms thus requiring broadening of ceftriaxone to cefepime • Adjust cefepime, vancomycin, meropenem, and aztreonam in patients with renal dysfunction • CT prior to lumbar puncture if: ○ Immunocompromised ○ History of CNS disease (mass lesion, stroke) ○ New onset seizures ○ Papilledema ○ Abnormal level of consciousness ○ Focal neurologic deficit • For infants and children with CSF shunts, vancomycin alone may be appropriate if GNRs are not identified on gram-stain
Penetrating trauma *S. aureus*, Coag-negative *Staphylococci*, Aerobic gram-negative bacilli (including *Pseudomonas*)	Preferred: **Cefepime** 50 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Non-life threatening PCN/cephalosporin allergy: **Meropenem** 40 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Life threatening penicillin allergy: **Aztreonam** 40 mg/kg/dose IV q6h (max: 2 g/dose) + **Vancomycin** IV		
Post neurosurgery Aerobic gram-negative bacilli (including *Pseudomonas*) *S. aureus*, Coag-negative *Staphylococci*	Preferred: **Cefepime** 50 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Non-life threatening PCN/cephalosporin allergy: **Meropenem** 40 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Life threatening penicillin allergy: **Aztreonam** 40 mg/kg/dose IV q6h (max: 2 g/dose) + **Vancomycin** IV		

Common Pathogens	Empiric Therapy	Duration of Therapy	Comments/Reference
Presence of CSF shunt Aerobic gram-negative bacilli (including *Pseudomonas*), *S. aureus*, Coag-negative *Staphylococci, Propionibacterium acnes*	Preferred: **Cefepime** 50 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Non-life threatening PCN/cephalosporin allergy: **Meropenem** 40 mg/kg/dose IV q8h (max: 2 g/dose) + **Vancomycin** IV Life threatening penicillin allergy: **Aztreonam** 40 mg/kg/dose IV q6h (max: 2 g/dose) + **Vancomycin** IV	7-21 days (plus)	• CSF shunt infections: Gold standard for infection clearance is removal of shunt. Prior to replacement of shunt, cultures should be negative for: ○ Coagulase negative Staphylococci + normal CSF findings: 3 days ○ Coagulase negative Staphylococci + abnormal CSF findings: 7 days ○ S. aureus: 10 days ○ Gram negative bacilli: 10-14 days (plus) • For infants and children with CSF shunts, vancomycin alone may be appropriate if GNRs are not identified on gram-stain • Adjust cefepime, vancomycin, meropenem, and aztreonam in patients with renal dysfunction

*These guidelines are not intended for use in neonatal patients who have been hospitalized beyond the initial postpartum time period.

Last revised: 03/2021	**Antimicrobial Stewardship Subcommittee:** N/A	**P&T:** 11/2011

References

	Citation
1	Tunkel AR et al. Practice Guidelines for the Management of Bacterial Meningitis. Clin Infect Dis. 2004. 39 (9), 1267-84.

Revision History

Date	Revision
03/2021	Updated vancomycin hyperlinks

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.